Why TeX Still Matters

Typesetting Club

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1 Introduction

Most documents today are written in word processors, yet many mathematicians, physicists and computer scientists still write their papers in LaTeX. This essay explains where TeX came from, why LaTeX was built on top of it, and why both are still in use almost fifty years later.

2 From Metal to Code

For five centuries, books were printed from metal type that skilled workers arranged by hand or by machine. Mathematics was the hardest text to set, because formulas mix symbols of many sizes on several levels. In the 1970s publishers moved to photographic and then digital typesetting, and the quality of mathematical books dropped sharply.

3 Knuth's TeX

Knuth (1979) described how disappointed he was when he saw the proofs of a new edition of his own book. He decided to write a program that would set mathematics as well as the best printers of the past, and spent almost ten years on it. The result was TeX, documented in full detail in *The TeXbook* (Knuth 1984).

TeX treats a page as boxes and glue: characters and formulas are boxes, and the spaces between them are glue that can stretch or shrink. The program chooses the line breaks for a whole paragraph at once, which is why its output looks so even.

4 LaTeX

TeX gives the author complete control, but it also asks the author to think about every detail of the layout. Leslie Lamport wrote LaTeX as a set of macros on top of TeX so that authors could describe the *structure* of a document, such as sections, lists and figures, and leave the layout to the system (Lamport 1994).

Today the most complete reference to LaTeX and its packages is Mittelbach and Fischer (2023), while beginners usually start with a free guide (Oetiker et al. 2021).

5 TeX Today

Thousands of packages extend LaTeX for chemistry, music, chess, linguistics and many other subjects. They are collected in one archive (CTAN Team 2026), and distributions such as TeX Live install them all at once. Because the core program has hardly changed since 1990, a document written forty years ago can still be compiled today (Knuth 1984; Lamport 1994).

References

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